

Oreste Muhirwa Gabo

Software Engineer – Mobile Platforms, Backend & Digital Systems

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Available to relocate independently to Rwanda for this position

Professional Summary

Software engineer experienced in delivering citizen-facing mobile services, secure backend integrations, and real-time communication features in a professional environment. I work across Android, iOS, APIs, data, security, and deployment, with particular attention to reliability on low-bandwidth networks and entry-level devices. A fast learner, I adapt quickly to unfamiliar technologies and operational needs, working effectively both independently and with technical and non-technical teams. I am interested in contributing this end-to-end engineering experience to MTN's customer-facing digital platforms and services.

Technical Skills

Mobile: Kotlin, Java, Jetpack Compose, SwiftUI, Kotlin Multiplatform, Flutter, Android

Backend and APIs: Spring Boot, Ktor, REST, Swagger/OpenAPI, Postman, k6, JWT, OAuth2, WebRTC, LiveKit

Messaging and data: MQTT, RabbitMQ, Redis, PostgreSQL, MySQL, SQLite, SQL, ETL and data migration

Cloud and delivery: Docker, Kubernetes, Helm, Cloud Run, GitHub Actions, CI/CD, production troubleshooting

Operations and observability: Linux, Bash, Zabbix, Prometheus, probes, resources, ingress, ConfigMaps, secrets

Networks and security: TCP/IP, DNS, TLS, advanced subnetting, Wireshark, load balancing, reverse proxies, OWASP/mobile security

Telecom fundamentals: Basic 3G/4G/5G concepts, latency, availability, service continuity, low-bandwidth operation

Additional languages: C++, Python, TypeScript, JavaScript

Professional Experience

Software Engineer – Digital Platforms & Systems

June 2023 – Present

CDAFAL / City of Mulhouse, France

- Develop and maintain a citizen-facing mobile platform providing secure access to municipal services, administrative documents, and support workflows.
- Build native mobile experiences with **Kotlin**, **Jetpack Compose**, and **SwiftUI**, translating operational requirements into clear and dependable user journeys.
- Work across mobile clients and **Spring Boot REST APIs**, using Postman and Swagger/OpenAPI to coordinate contracts, validation, authentication, error handling, and end-to-end behaviour.
- Integrated **LiveKit/WebRTC** to support secure real-time video consultations between citizens and public-service staff.
- Implement and maintain **JWT-secured API flows** following OWASP and mobile-security fundamentals for sensitive data, authentication, secure communication, input validation, and secrets.
- Improve application stability, responsiveness, and network behaviour for users with low bandwidth, limited data plans, entry-level smartphones, and constrained devices.
- Diagnose defects across mobile, API, data, network, and infrastructure layers using Linux/Bash tools, **Zabbix**, **Prometheus**, and **Wireshark**.
- Collaborate with technical colleagues and service stakeholders to clarify requirements, communicate constraints, review feedback, and turn operational needs into deliverable software.
- Take ownership of assigned work from analysis through implementation, testing, documentation, and follow-up, while escalating risks and dependencies when needed.
- Organise priorities across mobile, backend, and production concerns to meet agreed deadlines without losing sight of security, reliability, or user impact.
- Adapt rapidly across Kotlin, SwiftUI, Spring Boot, WebRTC, and data workflows as project needs evolve.
- Contribute to delivery through GitHub Actions, Docker, Kubernetes and Helm, including probes, resources, ingress, ConfigMaps, secrets, and environment-aware configuration.
- Balance usability, security, performance, and service continuity for real users depending on public digital services.

Web Developer Intern

January 2023 – June 2023

CDAFAL / City of Mulhouse, France

- Analysed manual registration, fee-tracking, examination, language-course, and student-record workflows before designing their digital replacement.
- Built internal automation tools and a responsive **Vue.js** interface backed by **Spring Boot** services.
- Developed a custom **C++ ETL** tool to migrate, validate, and process fragmented operational data.
- Replaced paper, spreadsheet, and legacy database processes with a centralised application and structured relational data.
- Tested features, corrected defects, documented the delivered system, and adapted the implementation from user feedback.

Selected Engineering Projects

SchoolBridge V2: Kotlin and Spring Boot education platform with role-based workflows, safety alerts, and MQTT messaging.

Kaze: Kotlin Multiplatform product with Ktor, PostgreSQL, OpenAPI, JWT/OAuth2, Docker, Cloud Run, and CI/CD.

SkyWatch Radar: Java/Spring Boot real-time event-processing prototype using MQTT with latency and jitter measurement.

Education

Computer Science and Mobility studies – Bac+4 level completed

2024 – 2025

Université de Haute-Alsace, Mulhouse, France

Four completed years of higher education after secondary school. In the French system, **Bac+4** represents 240 ECTS credits and completion of the first year of the Master's study cycle; it describes a completed study level, not a Master's degree. Coursework included software engineering, application security, systems, networks, databases, cloud computing, and mobile development.

Bachelor's degree in Computer Science (Licence Informatique) – Université de Haute-Alsace

Completed